

Solar Elevation Estimation from Single Images Using Convolutional Neural Networks for Multi-Image Geolocation

[Alex Nowlan]

Department of Computer Science and Software Engineering
University of Canterbury, Christchurch, New Zealand
[ano58@uclive.ac.nz]

Abstract—[TODO: Write the abstract last. Cover: (1) problem statement – estimating geographic location from consumer photographs without GPS metadata; (2) your approach – fine-tuning an ImageNet-pretrained CNN (ResNet/VGG/etc.) to regress solar elevation angle from a single image, combined with an astronomical sun-position model to infer latitude/longitude from a set of temporally spaced, co-located images; (3) key quantitative result, e.g. “median location error of X km on our test set”; (4) significance – passive, metadata-free geolocation applicable to archival or forensic photography.]

Index Terms—solar elevation, geolocation, convolutional neural network, transfer learning, sun position model, ImageNet, image forensics

I. INTRODUCTION

[TODO: Write the introduction following the five-paragraph structure above. Be specific: name the CNN backbone you chose and why (e.g. ResNet-50 – good accuracy/speed trade-off, widely benchmarked). Mention that timestamps must be available in EXIF or supplied manually.]

The automatic determination of where a photograph was taken — without relying on embedded GPS metadata — is an open problem in computer vision with applications in digital forensics [7], investigative journalism, and autonomous navigation. [TODO: Expand this opening paragraph and add real citations.]

II. RELATED WORK

A. Image-Based Geolocation

[TODO: Discuss landmark / scene-matching approaches (e.g. IM2GPS [3], PlaNet [4]). Note they require large retrieval databases and fail on featureless outdoor scenes.]

B. Solar Cues in Computer Vision

[TODO: Cover work that uses shadows or sky appearance to estimate sun direction (e.g. hold-geoffroy2017 deep sky model, lalonde2012 outdoor illumination). Contrast with your regression-based, end-to-end CNN approach.]

C. Transfer Learning for Regression

[TODO: Briefly explain why fine-tuning an ImageNet-pretrained network for a regression target (elevation angle in degrees) is sound practice. Reference a general transfer learning paper and one domain-specific example.]



Fig. 1: System pipeline. A CNN estimates solar elevation $\hat{\theta}$ from each input image. Given timestamps, the sun-position model inverts the elevation to candidate (latitude, longitude) loci. Multiple images constrain the solution to a unique geographic location.

D. Astronomical Sun-Position Models

[TODO: Cite the algorithm you use (e.g. NREL Solar Position Algorithm, Grena 2012, or pysolar). Note accuracy (sub-0.01° for SPA).]

III. METHODOLOGY

A. System Overview

[TODO: Insert your pipeline diagram (Fig. 1) and write 2–3 sentences describing the end-to-end flow.]

B. CNN Architecture and Transfer Learning

[TODO: State which backbone you used (e.g. ResNet-50 pretrained on ImageNet-1K [1]). Describe the modification: replace the final classification head with a single-neuron regression head outputting elevation $\theta \in [-90, 90]$. Discuss:

- Layer freezing strategy (which layers were frozen / fine-tuned).
- Loss function (MSE or MAE for regression).
- Activation on output neuron (none, or tanh scaled to $[-90, 90]$).

The backbone network f_ϕ maps an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ to a predicted solar elevation angle:

$$\hat{\theta} = f_\phi(\mathbf{I}), \quad \hat{\theta} \in [-90, 90]. \quad (1)$$

[TODO: Explain the training objective, e.g. $\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (\hat{\theta}_i - \theta_i)^2$.]

C. Dataset and Ground-Truth Labels

[TODO: Describe:

- Where your training images came from (Flickr, personal collection, public dataset). State the total number and train/val/test split.
- How ground-truth elevation angles were computed from EXIF timestamps and known GPS coordinates using your sun-position library.
- Any data augmentation applied (random crop, colour jitter, etc.).
- Class / angle distribution (histogram would be great as a figure).

]

D. Sun–Earth Geometric Model

[TODO: Explain the astronomical model you use to compute solar elevation θ given observer latitude ϕ , longitude λ , date, and UTC time t :

$$\theta = g(\phi, \lambda, t), \quad (2)$$

where $g(\cdot)$ is the Solar Position Algorithm (or equivalent). Cite the algorithm. Note the precision achievable (< 0.01 for SPA).]

E. Geolocation via Multi-Image Constraint Solving

[TODO: This is the novel contribution of your pipeline. Explain:

- For a single image with timestamp t_k and predicted elevation $\hat{\theta}_k$, equation (2) yields a locus of (ϕ, λ) pairs (a curve on the Earth’s surface).
- Two or more such curves (from different times) intersect at the true location.
- How you solve for the intersection (e.g. least-squares optimisation, grid search, or Levenberg–Marquardt on the residual $\sum_k (g(\phi, \lambda, t_k) - \hat{\theta}_k)^2$).
- Any ambiguities (e.g. hemisphere ambiguity) and how you handle them.

A small diagram showing the intersecting loci on a map would strengthen this sub-section.]

The geolocation estimate $(\hat{\phi}, \hat{\lambda})$ minimises:

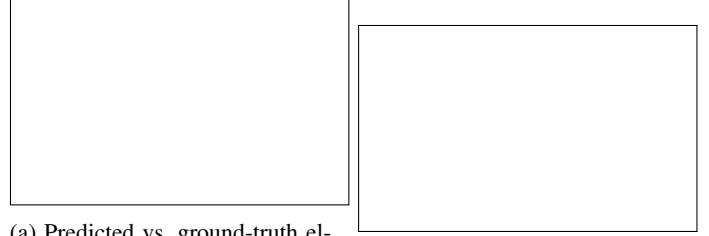
$$(\hat{\phi}, \hat{\lambda}) = \arg \min_{\phi, \lambda} \sum_{k=1}^K (g(\phi, \lambda, t_k) - \hat{\theta}_k)^2. \quad (3)$$

F. Implementation Details

[TODO: List hyperparameters and software stack concisely (a table is clean): backbone, input resolution, batch size, optimiser (Adam, SGD), learning rate schedule, number of epochs, hardware (GPU model), framework (PyTorch / TensorFlow), sun-position library (pysolar, pvlib, or custom). This lets readers reproduce your work.]

TABLE I: Implementation hyperparameters

Parameter	Value
Backbone	ResNet-50 (ImageNet-1K pretrained)
Input resolution	224×224
Batch size	[TODO: X]
Optimiser	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Initial LR	[TODO: X]
LR schedule	[TODO: e.g. cosine decay]
Epochs	[TODO: X]
Sun-position lib	[TODO: e.g. pysolar 0.9]
Framework	[TODO: PyTorch X.X]
Hardware	[TODO: e.g. NVIDIA GTX 1080 Ti]



(a) Predicted vs. ground-truth elevation.

(b) Residual histogram.

Fig. 2: CNN elevation estimation performance on the test set. [TODO: Fill in actual MAE/RMSE numbers in the caption or text.]

IV. EXPERIMENTS AND RESULTS

A. Evaluation Metrics

[TODO: Define all metrics you report:

- Elevation MAE / RMSE (degrees): primary CNN accuracy metric.
- Geolocation error (km): great-circle distance between predicted and ground-truth coordinates. State whether you report mean, median, or top-X% accuracy.

]

B. CNN Solar Elevation Estimation

[TODO: Report test-set MAE and RMSE. Compare with:

- A naive baseline (e.g. predict mean training elevation for all images).
- Other backbone choices if you ablated them.

Include a scatter plot of predicted vs. ground-truth elevation angles (Fig. 2a) and a residual distribution histogram.]

TABLE II: Elevation estimation accuracy (test set)

Model	MAE (°)	RMSE (°)
Mean baseline	[TODO:]	[TODO:]
[TODO: Backbone A]	[TODO:]	[TODO:]
[TODO: Backbone B]	[TODO:]	[TODO:]
Ours (final)	[TODO:]	[TODO:]

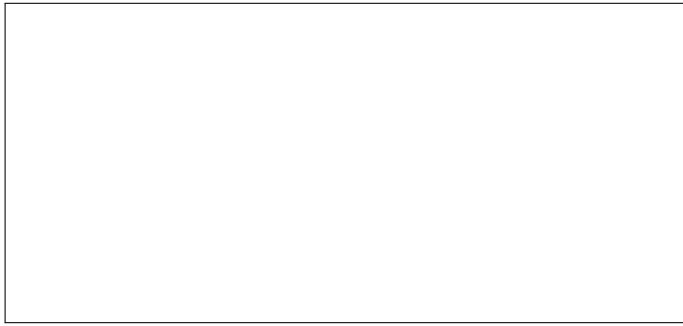


Fig. 3: Example geolocation results. $\star$ = ground truth; $\bullet$ = predicted location. [TODO: Replace placeholder with an actual map figure.]

C. Geolocation Accuracy

[TODO: Report geolocation results. Suggested structure:

- Table or figure showing median / mean error (km) as a function of the number of input images K (e.g. $K = 2, 3, 4, 6$).
- A world map or local map showing predicted vs. true locations for representative test sequences (Fig. 3).
- Discuss upper-bound performance using ground-truth elevations instead of CNN estimates, to separate CNN error from model error.

D. Ablation Study

[TODO: Optional but strongly recommended. Test the effect of:

- Number of images K on geolocation error.
- Degree of fine-tuning (head only vs. last block vs. full network).
- Input resolution.
- Time-of-day distribution in training data.

E. Qualitative Analysis

[TODO: Show 2–3 interesting examples (success cases and failure cases). For each, display the image, the predicted elevation, and the resulting location estimate. Discuss what visual cues the model appears to be using (shadow length/direction, sky colour gradient, haze) using a Grad-CAM or similar visualisation if possible.]

V. DISCUSSION

[TODO: Address the following points:

- What worked well? Which conditions (clear sky, strong shadows, golden hour) produce the most accurate elevation estimates?
- What failed and why? Overcast skies, indoor/artificial lighting, extreme latitudes, near-solstice dates (low sun-movement rate), timestamp errors.

- Error budget: How much of the final geolocation error is attributable to CNN elevation error vs. inherent geometric ambiguity?
- Practical considerations: Minimum number of images needed for reliable geolocation; effect of image spacing in time; hemisphere ambiguity.
- Comparison to GPS accuracy: Put your error (km) in context.

VI. CONCLUSION

[TODO: Write 2–3 paragraphs:

- 1) Restate the problem and your key contributions (1–2 sentences each, do not copy the abstract verbatim).
- 2) Your best result in one sentence (e.g. “With four co-located images our system achieves a median geolocation error of X km without any landmark database or prior map knowledge.”).
- 3) Future work: fine-tuning on more diverse data, adding moon/star cues for night images, uncertainty estimation, real-time deployment.

ACKNOWLEDGEMENTS

The author thanks Richard Green for project supervision and feedback. [TODO: Add any other acknowledgements (dataset providers, compute resources, etc.)]

REFERENCES

- [1] O. Russakovsky *et al.*, “ImageNet Large Scale Visual Recognition Challenge,” *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, 2015.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE CVPR*, pp. 770–778, 2016.
- [3] J. Hays and A. A. Efros, “IM2GPS: estimating geographic information from a single image,” in *Proc. IEEE CVPR*, 2008.
- [4] T. Weyand, I. Kostrikov, and J. Philbin, “PlaNet – photo geolocation with convolutional neural networks,” in *Proc. ECCV*, 2016.
- [5] I. Reda and A. Andreas, “Solar position algorithm for solar radiation applications,” *Solar Energy*, vol. 76, no. 5, pp. 577–589, 2004.
- [6] J.-F. Lalonde, A. A. Efros, and S. G. Narasimhan, “Estimating the natural illumination conditions from a single outdoor image,” *Int. J. Comput. Vis.*, vol. 98, no. 2, pp. 123–145, 2012.
- [7] [TODO: Add a relevant image forensics / provenance citation.]